22 Ekim 2021

* Balance following equations.

(1) $Al(OH)_3 + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2O$

$2\,Al(OH)_3 + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2O$

$\begin{pmatrix} 2\,Al \\ 10\,O \\ 8\,H \\ S \end{pmatrix} \longrightarrow \begin{pmatrix} 2\,Al \\ 13\,O \\ 2\,H \\ 3\,S \end{pmatrix}$

$2\,Al(OH)_3 + 3\,H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2O$

$\begin{pmatrix} 18\,O \\ 12\,H \end{pmatrix} \longrightarrow \begin{pmatrix} 13\,O \\ 2\,H \end{pmatrix}$

$2\,Al(OH)_3 + 3\,H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 6\,H_2O$

$\begin{pmatrix} 12\,H \\ 18\,O \end{pmatrix} \longrightarrow \begin{pmatrix} 12\,H \\ 18\,O \end{pmatrix}$ ✓

(2) $Cu + HNO_3 \longrightarrow Cu(NO_3)_2 + NO_2 + H_2O$

$\begin{pmatrix} Cu \\ H \\ N \\ 3\,O \end{pmatrix} \longrightarrow \begin{pmatrix} Cu \\ 2\,H \\ 3\,N \\ 9\,O \end{pmatrix}$

$Cu + 2\,HNO_3 \longrightarrow Cu(NO_3)_2 + NO_2 + H_2O$

$\begin{pmatrix} Cu \\ 2\,H \\ 2\,N \\ 6\,O \end{pmatrix} \longrightarrow \begin{pmatrix} Cu \\ 2\,H \\ 3\,N \\ 9\,O \end{pmatrix}$ H ✓ N X

$Cu + 4\,HNO_3 \longrightarrow Cu(NO_3)_2 + 2\,NO_2 + H_2O$

$$Cu + 4\,HNO_3 \longrightarrow Cu(NO_3)_2 + 2\,NO_2 + H_2O$$

$$\begin{pmatrix} Cu \\ 4H \\ 4N \\ 12O \end{pmatrix} \longrightarrow \begin{pmatrix} Cu \\ 2H \\ 4N \\ 11O \end{pmatrix}$$

to balance H

$$Cu + 4\,HNO_3 \longrightarrow Cu(NO_3)_2 + 2\,NO_2 + 2\,H_2O$$

$$\begin{pmatrix} 4H \\ 4N \\ 12O \end{pmatrix} \longrightarrow \begin{pmatrix} 4H \\ 4N \\ 12O \end{pmatrix}$$ ✓

(3) $C_5H_{10}O_2(\ell) + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$

$$C_5H_{10}O_2 + O_2 \longrightarrow 5\,CO_2 + H_2O$$

$$\begin{matrix} 5C \\ 10H \\ 4O \end{matrix} \longrightarrow \begin{matrix} 5C \\ 2H \\ 11O \end{matrix}$$

to balance H

$$C_5H_{10}O_2 + O_2 \longrightarrow 5\,CO_2 + 5\,H_2O$$

$$\begin{matrix} 5C \\ 10H \\ 4O \end{matrix} \longrightarrow \begin{matrix} 5C \\ 10H \\ 15O \end{matrix}$$ ✗

$$2\,C_5H_{10}O_2 + O_2 \longrightarrow 10\,CO_2 + 10\,H_2O$$

$$\begin{matrix} 10C \\ 20H \\ 4O + ? \end{matrix} \longrightarrow \begin{matrix} 10C \\ 20H \\ 30O \end{matrix}$$

$4 + 26 = 30$

$$2\,C_5H_{10}O_2 + 13\,O_2 \longrightarrow 10\,CO_2 + 10\,H_2O$$ ✓

* 5.325 g methyl benzoate contains 3.758 g C
0.316 g H
1.251 g O

what is the empirical formula?

$3.758\ g\ C \times \left(\frac{1\ mol}{12\ g}\right) = 0.313167 \;/\; 0.31287 = 1.00$

$0.316\ g\ H \times \left(\frac{1\ mol}{1.01\ g}\right) = 0.31287 \;/\; 0.31287 = 1.00$

$1.251\ g\ O \times \left(\frac{1\ mol}{16\ g\ O}\right) = 0.0782 \;/\; 0.31287 = 0.25$

$C_x H_x O_{0.25x}$

$x \cdot \underbrace{C_4H_4O}_{68\ g} \rightarrow 5.325\ g$

$x = 0.0783\ mol$

↳ so it contains
3.758 g C
0.313 g H
1.252 g O

* What mass of O_2 is consumed in the combustion of 1.00 g C_3H_8 (propane)?

$$C_3H_8 + O_2 \longrightarrow CO_2 + H_2O$$

$$C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O \quad \checkmark$$

$$1.00\ g\ C_3H_8 \times \left(\frac{1\ mol\ C_3H_8}{44\ g\ C_3H_8}\right) = 0.022727\ mol$$

$$0.022727\ mol\ C_3H_8 \times \left(\frac{5\ mol\ O_2}{1\ mol\ C_3H_8}\right) \times \left(\frac{32\ g\ O_2}{1\ mol\ O_2}\right) = 3.63\ g\ O_2$$

* Determine limiting reactant for 24 mol methanol and 15 mol of O_2.

$$2CH_3OH + 3O_2 \longrightarrow 2CO_2 + 4H_2O$$

$$\begin{pmatrix} 2C \\ 8H \\ 8O \end{pmatrix} \rightarrow \begin{pmatrix} 2C \\ 8H \\ 8O \end{pmatrix} \quad \checkmark$$

$2CH_3OH + 3O_2 \rightarrow 2CO_2 + 4H_2O$

24 mol 36 mol → X limiting reactant O_2

20 mol 15 mol → ✓

$2CH_3OH + 3O_2 \longrightarrow$

24 mol 15 mol

− 10 mol − 15 mol

14 mol CH_3OH excess reactant.

* If 3 g Ti metal is reacted with 6 g of chlorine gas ($Cl_2(g)$) to form 7.7 g of $TiCl_4$. what is the percent yield?

$$Ti(s) + 2Cl_2(g) \rightarrow TiCl_4(\ell)$$ Balanced ✓

$$3\ g\ Ti \times \left(\frac{1\ mol\ Ti}{47.86\ g\ Ti}\right) = 0.063\ mol\ Ti$$

$$6\ g\ Cl_2 \times \left(\frac{1\ mol\ Cl_2}{71\ g\ Cl_2}\right) = 0.0845\ mol\ Cl_2$$

$$\underbrace{Ti}_{0.063\ mol} + \underbrace{2Cl_2}_{0.126\ mol} \rightarrow$$

$\rightarrow$ X Cl_2 is limiting reactant

$$\underbrace{Ti}_{0.042\ mol} + \underbrace{2Cl_2}_{0.0845\ mol} \rightarrow \underbrace{TiCl_4}_{0.042\ mol\ TiCl_4\ forms}$$

theoritical yield $0.042\ mol\ TiCl_4 \times \left(\frac{189.86\ g\ TiCl_4}{1\ mol\ TiCl_4}\right) = 7.9$ Ti

percent yield $\frac{7.7\ g\ TiCl_4}{7.97\ g\ TiCl_4} \times 100\% = 96.6\ \%$